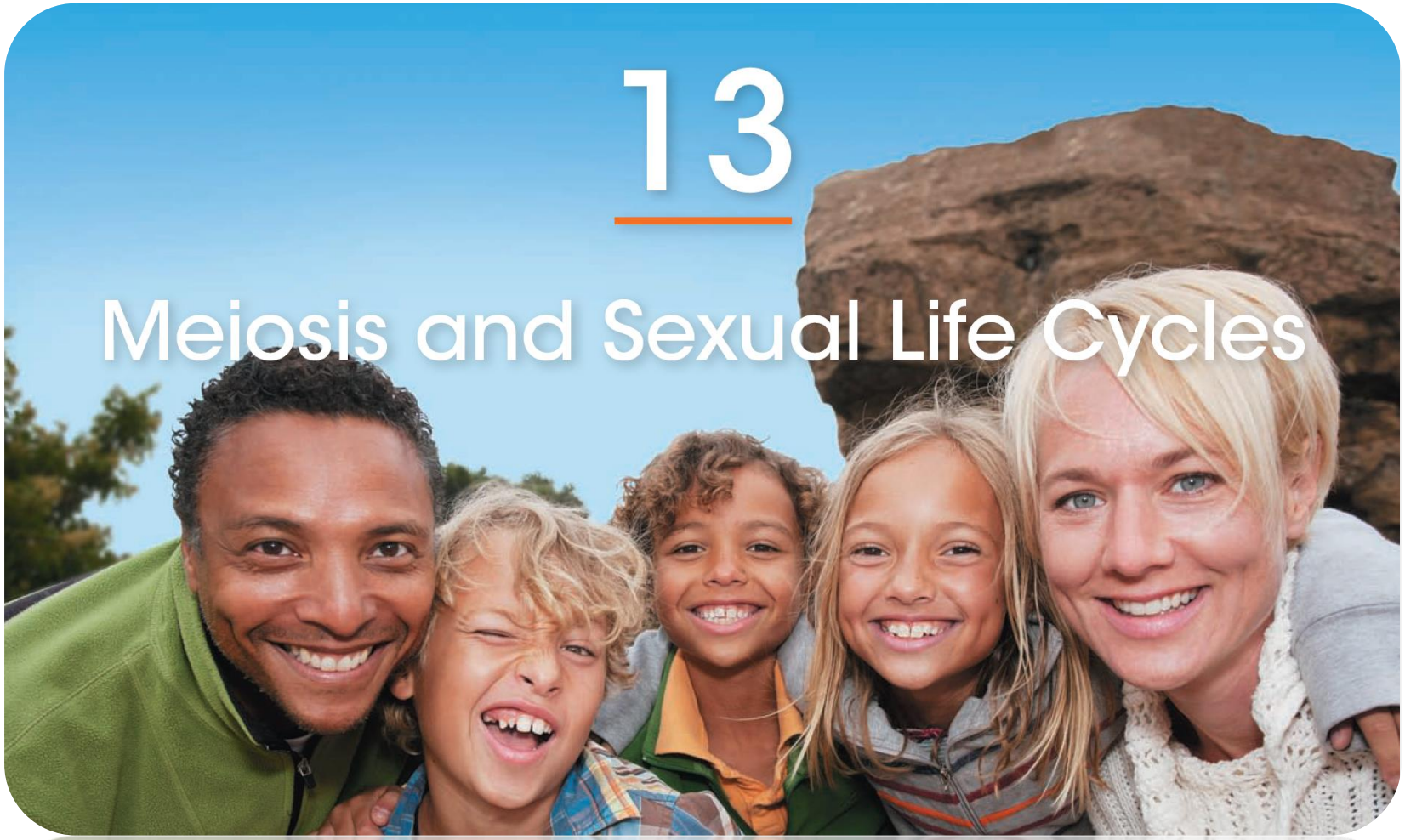
LECTURE-13

Meiosis

INDRAPRASTHA INSTITUTE *of*
INFORMATION TECHNOLOGY
DELHI

The cell cycle



KEY CONCEPTS

13.1 Offspring acquire genes from parents by inheriting chromosomes

13.2 Fertilization and meiosis alternate in sexual life cycles

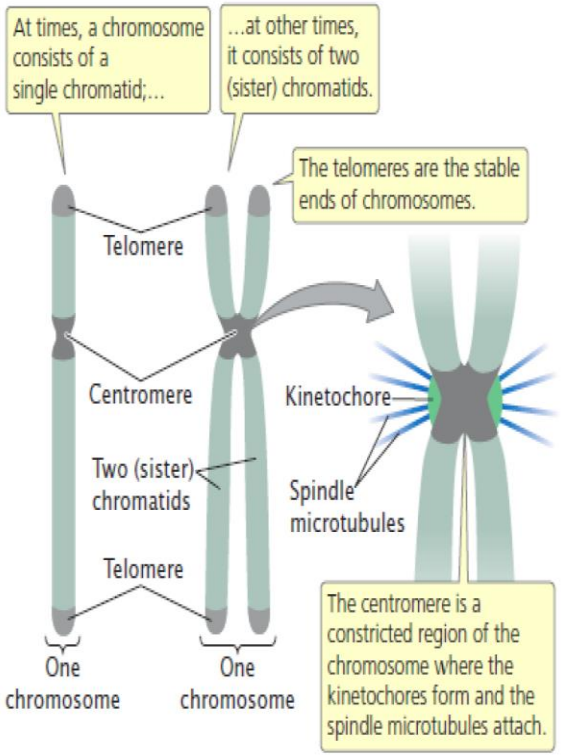
13.3 Meiosis reduces the number of chromosome sets from diploid to haploid

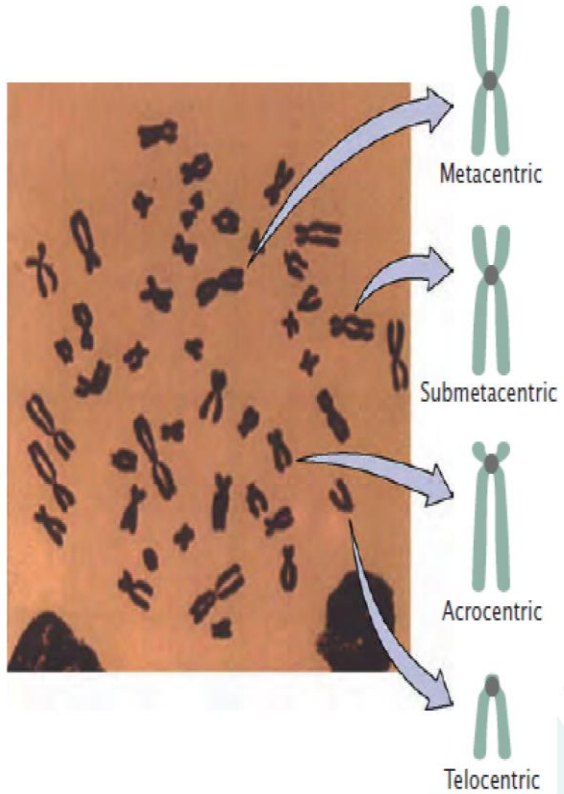
13.4 Genetic variation produced in sexual life cycles contributes to evolution

Some terminologies

- In most eukaryotic cells, there are **two sets of chromosomes**.
- One set is inherited from the male parent and the other from the female parent.
- Each chromosome in one set has a corresponding chromosome in the other set, together constituting a **homologous pair.**
- Human cells, for example, have 46 chromosomes, constituting 23 homologous pairs.
- The two chromosomes of a **homologous pair are usually alike** in structure and size, and each carries genetic information for the same set of hereditary characteristics.

CHROMOSOME = DNA + HISTONE PROTEINS

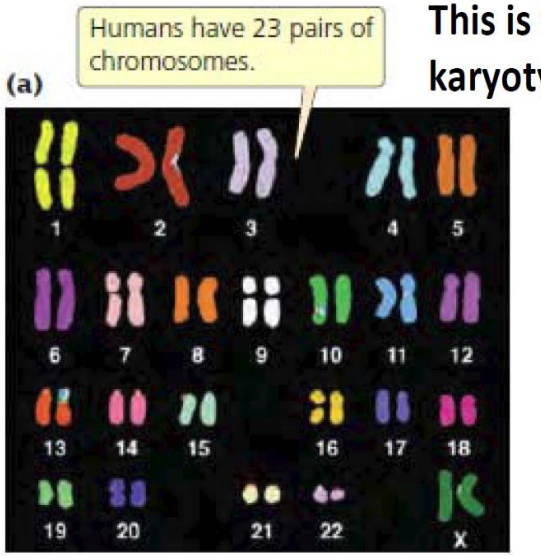




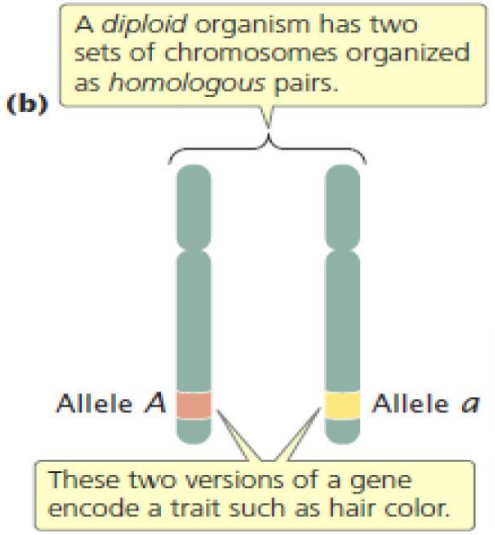
NOTE THIS !
ONLY ONE CHROMOSOME
BUT TWO SISTER CHROMATIDS; i.e, DNA is doubled.

Diploid, alleles, and summary of important terms

- Most cells carry two sets of genetic information; these cells are **diploid.**
- Cells with a single set of chromosomes are **haploid.** A haploid cell has only one copy of each gene.
- Reproductive cells (such as eggs, sperm, and spores) a single set of chromosomes.



This is the karyotype



Summary of important genetic terms

Term	Definition
Gene	A genetic factor (region of DNA) that helps determine a characteristic
Allele	One of two or more alternate forms of a gene
Locus	Specific place on a chromosome occupied by an allele
Genotype	Set of alleles possessed by an individual organism
Heterozygote	An individual organism possessing two different alleles at a locus
Homozygote	An individual organism possessing two of the same alleles at a locus
Phenotype or trait	The appearance or manifestation of a character
Character or characteristic	An attribute or feature

Genetics—Heredity and variations

- Living organisms are distinguished by their **ability to reproduce their own kind**.
- Offspring resemble their parents more than they resemble less closely related individuals of the same species.
- The transmission of traits from one generation to the next is called **heredity** or inheritance.
- Offspring differ somewhat from their parents and siblings, which demonstrates **variation**.
- Farmers have bred plants and animals for desired traits for thousands of years, but the mechanisms of heredity and variation eluded biologists until the *development of genetics* in the 20th century.
- **Genetics** is the scientific study of heredity and hereditary variation.
  - *Genetics has revolutionized medicine and agriculture.*
  - *Our acquisition and analysis of the genome sequences of many species has taught us a great deal about evolution at the molecular level.*
  - *Our ability to manipulate the genetic material, DNA, has raised some thorny social and ethical questions.*

Genes and DNA

- Parents endow their offspring with coded information in the form of **genes**.
  - ✓ *Your genome is made up of the genes that you inherited from your mother and your father.*
- Genes program specific traits that emerge as we develop from fertilized eggs into adults.
- **Genes are segments of DNA**. Genetic information is transmitted as specific sequences of the four deoxyribonucleotides in DNA.
  - ✓ ***This process is analogous to the symbolic information of language, in which words and sentences are translated into mental images.***
  - ✓ *Cells translate genetic "sentences" into freckles and other features that have no resemblance to genes*.
- Most genes program cells to synthesize specific enzymes and other proteins whose cumulative action produces an organism's inherited traits.
- **The transmission of hereditary traits has its molecular basis in the precise replication of DNA.**
  - ✓ *The replication of DNA produces copies of genes that can be passed from parents to offspring*.

Gametes, zygotes and somatic cells

- **In plants and animals, reproductive cells called gametes transmit genes from one generation to the next.**

- After **fertilization (fusion of a sperm cell and an ovum),** genes from both parents are present in the **nucleus of the fertilized egg, or zygote**.

- *Almost all* the DNA in a eukaryotic cell is subdivided into chromosomes in the nucleus.
  - ✓ *Tiny amounts of DNA are also found in the mitochondria and chloroplasts.*

- Every living species has a characteristic number of chromosomes.
  - ✓ *Humans have 46 chromosomes in their **somatic cells** – cells other than the gametes and their precursors.*

- **Each chromosome consists of a single DNA molecule associated with various proteins.**

- Each chromosome has hundreds or thousands of genes, each at **a specific location**, or **locus,** along the length of the chromosome.

Sexual and asexual reproduction

Like begets like, more or less: a comparison of asexual and sexual reproduction.

- Only organisms that reproduce asexually can produce offspring that are exact copies of them.

- In **asexual reproduction,** a single individual is the sole parent to donate genes to its offspring.

  - ✓ *Single-celled eukaryotes can reproduce asexually by mitotic cell division to produce two genetically identical daughter cells.*
  - ✓ *An individual that reproduces asexually gives rise to a **clone**, a group of genetically identical individuals.*
  - ✓ *Members of a clone may be genetically different as a result of mutation.*

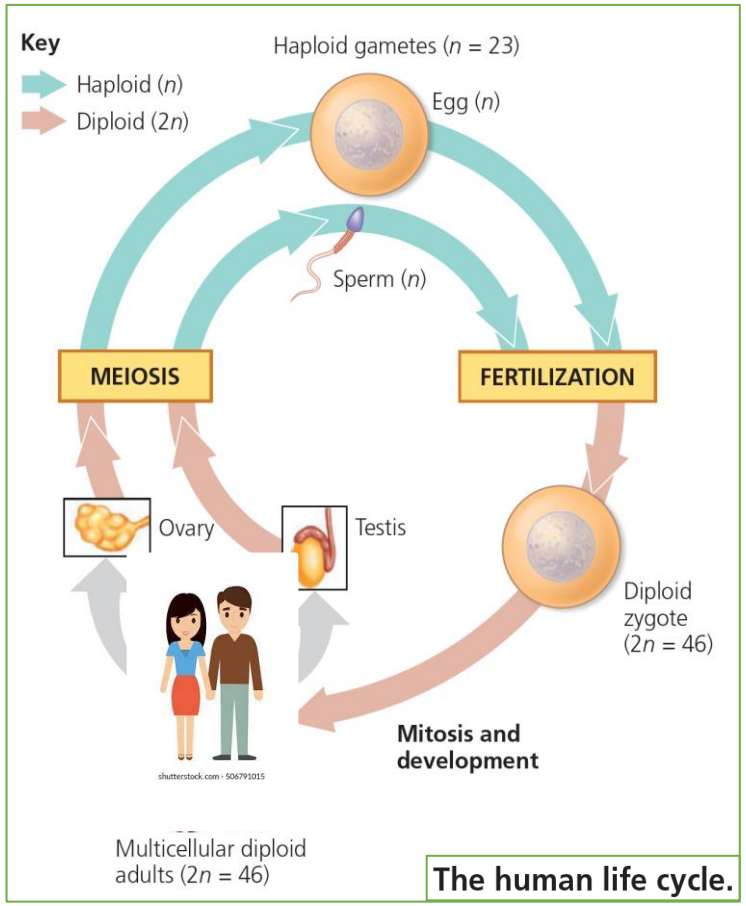
- In **sexual reproduction,** two parents produce offspring that have unique combinations of genes inherited from the two parents.

  - ✓ ***Unlike a clone, offspring produced by sexual reproduction vary genetically from their siblings and their parents.***

Fertilization and meiosis alternate in sexual life cycles

❖ **A <u>life cycle</u> is the generation-to-generation sequence of stages in the reproductive history of an organism.**

✓ *A life cycle starts at the conception of an organism and continues until the organism produces its own offspring.*



The human life cycle.

❖ In humans, each somatic cell has 46 chromosomes.

❖ During mitosis, the chromosomes become condensed and are visible under a light microscope.

✓ *At this point, chromosomes can be distinguished by their size, position of the centromere, and pattern of staining with certain dyes.*

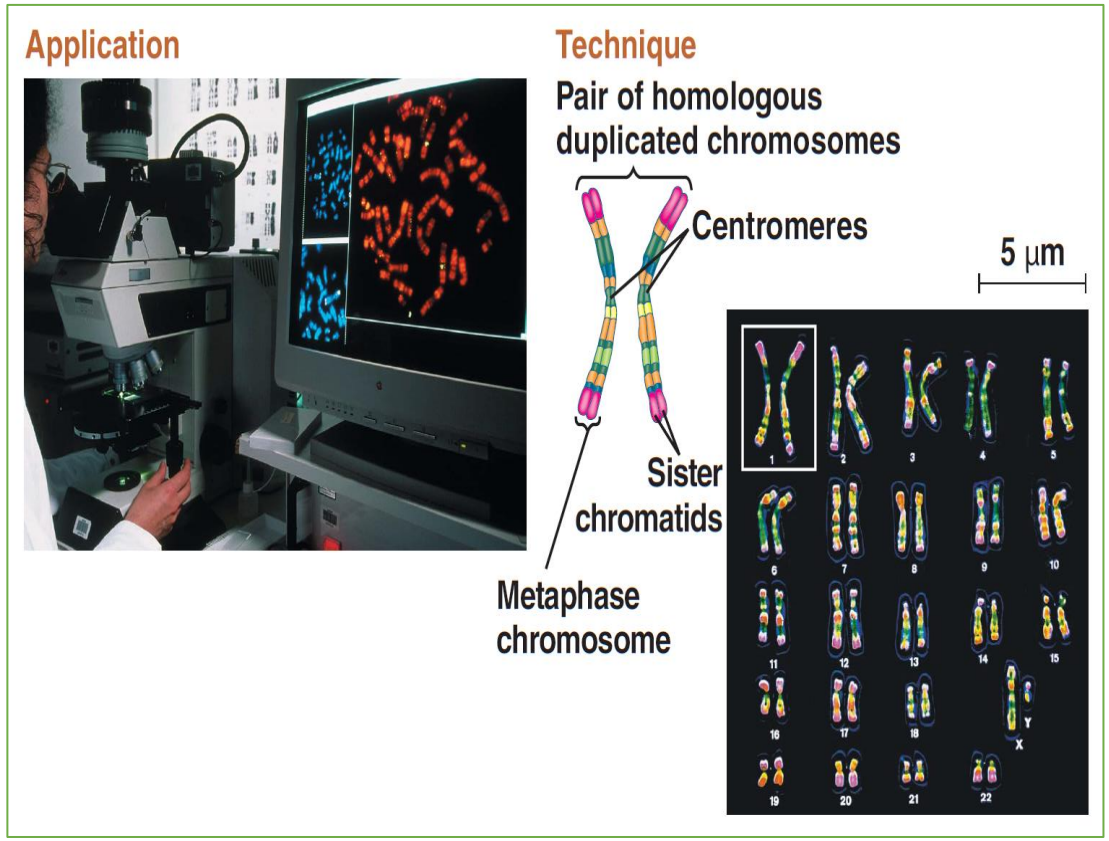
❖ Images of the 46 human chromosomes can be arranged in pairs in order of size to produce a **karyotype** display.

✓ *The two chromosomes in a pair have the same length, centromere position, and staining pattern.*

✓ *These **homologous chromosome** pairs carry genes that control the same inherited characters.*

Sex chromosomes and Karyotypes

- Two distinct **<u>sex chromosomes</u>,** the X and the Y, are an exception to the general pattern of homologous chromosomes in human somatic cells.
  - ✓ *The pattern of inheritance of the sex chromosomes determines an individual's sex.*
  - ✓ *Human females have a homologous pair of X chromosomes (XX); males have one X and one Y chromosome (XY).*
- Most of the genes carried on the X chromosome are not found on the tiny Y chromosome, which has genes that are not found on the X chromosome.
  - ✓ Only small parts of the X and Y chromosomes are homologous.



Application A karyotype is a display of condensed chromosomes arranged in pairs. Karyotyping can be used to screen for defective chromosomes or abnormal numbers of chromosomes associated with certain congenital disorders, such as Down syndrome.

Human chromosome, haploid and diploid

- The other 22 pairs of chromosomes are called **autosomes**.

- The occurrence of homologous pairs of chromosomes is a consequence of sexual reproduction.

  - ✓ *We inherit one chromosome of each homologous pair from each parent.*

  - ✓ *The 46 chromosomes in each somatic cell are two sets of 23, a maternal set (from your mother) and a paternal set (from your father).*

- The number of chromosomes in a single set is represented by *n*.

- Any cell with two sets of chromosomes is called a **diploid cell** and has a diploid number of chromosomes, abbreviated as 2*n*.

- Sperm cells or ova (gametes) have only one set of chromosomes—*22 autosomes and an X (in an ovum) or 22 autosomes and an X or a Y (in a sperm cell).*

Human chromosome, haploid and diploid

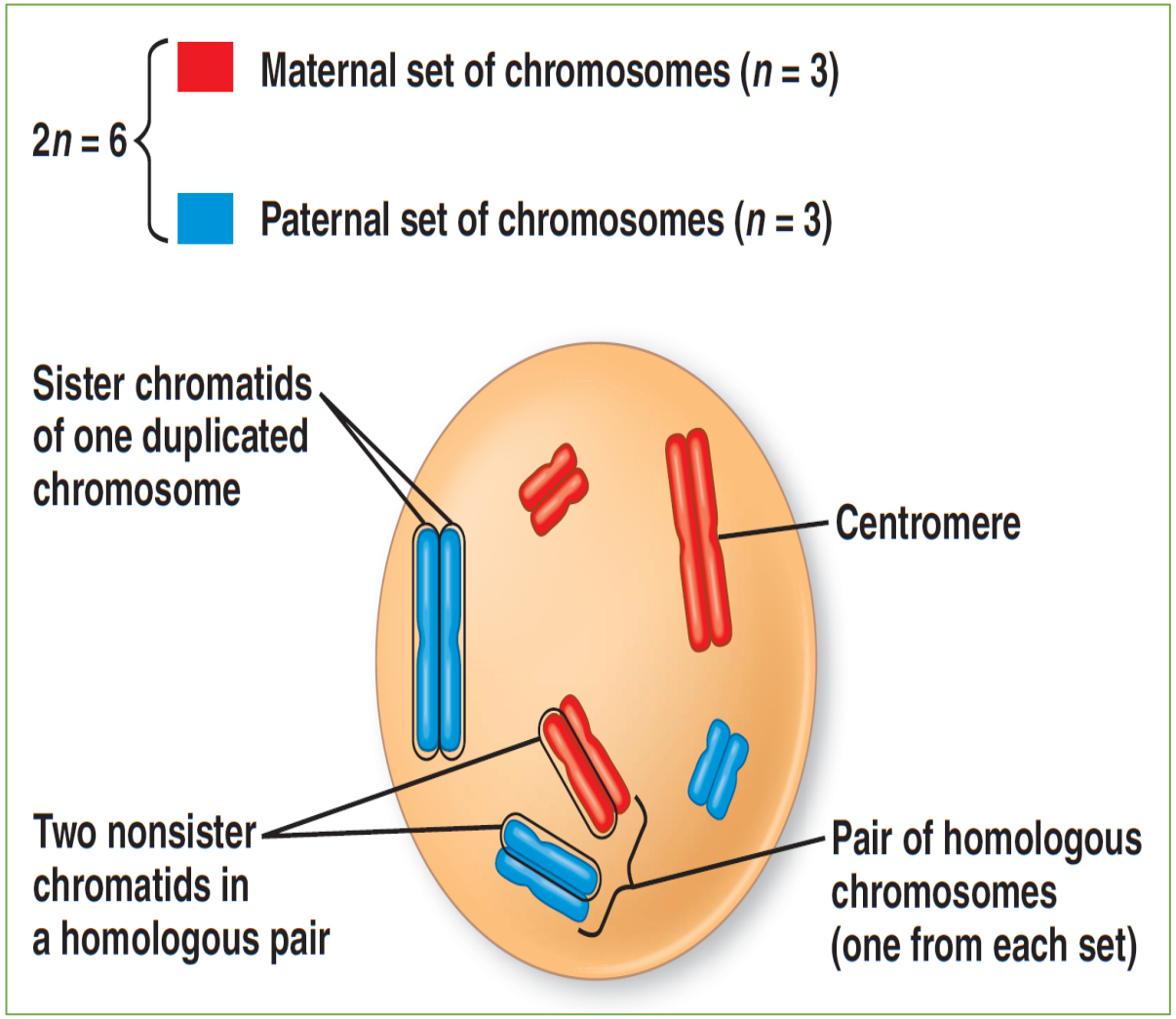
- ✓ *A cell with a single chromosome set is a* ***haploid cell****, abbreviated as n. This is a haploid number.*

❖ Any sexually reproducing species has characteristic haploid and diploid numbers of chromosomes.

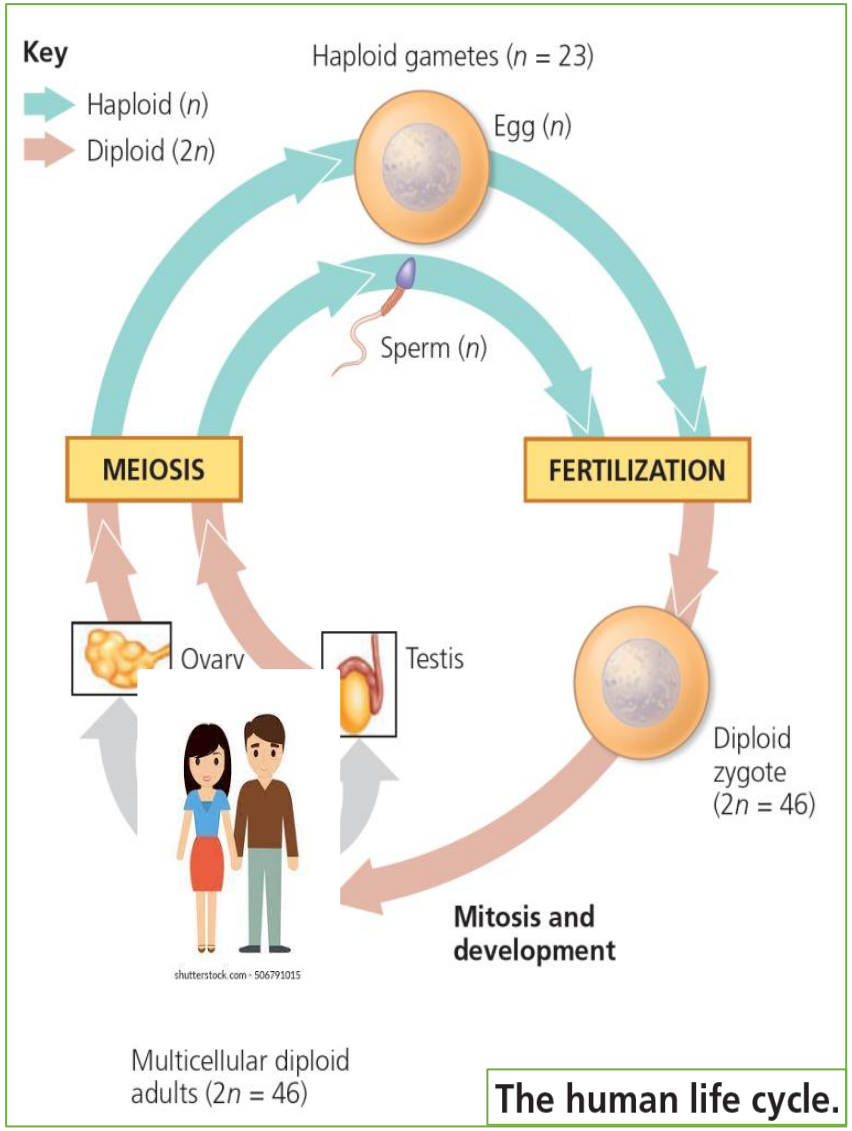
- ✓ *For humans, the haploid number (only one set out of two) of chromosomes is 23 (n = 23), and the diploid number (two sets, maternal and paternal) is 46 (2n = 46). In humans, we have 23* ***pairs*** of chromosomes.
- ✓ *In the fruit fly, Drosophila, 2n = 8 and n = 4.*
- ✓ *In the domestic dog, 2n = 78 and n = 39.*

❖ In a cell in which DNA synthesis has occurred, all the chromosomes are duplicated. Each duplicated chromosome consists of two identical sister chromatids.

❖ It is crucial to understand the differences among homologous chromosomes, sister chromatids, non sister chromatids, and chromosome sets.



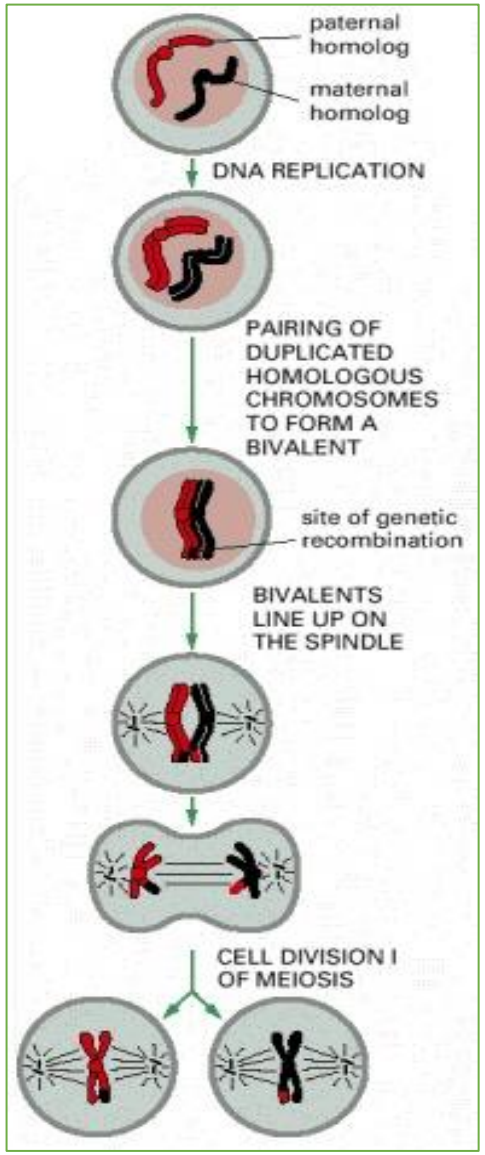
Behavior of chromosome sets in the human life cycle



The human life cycle.

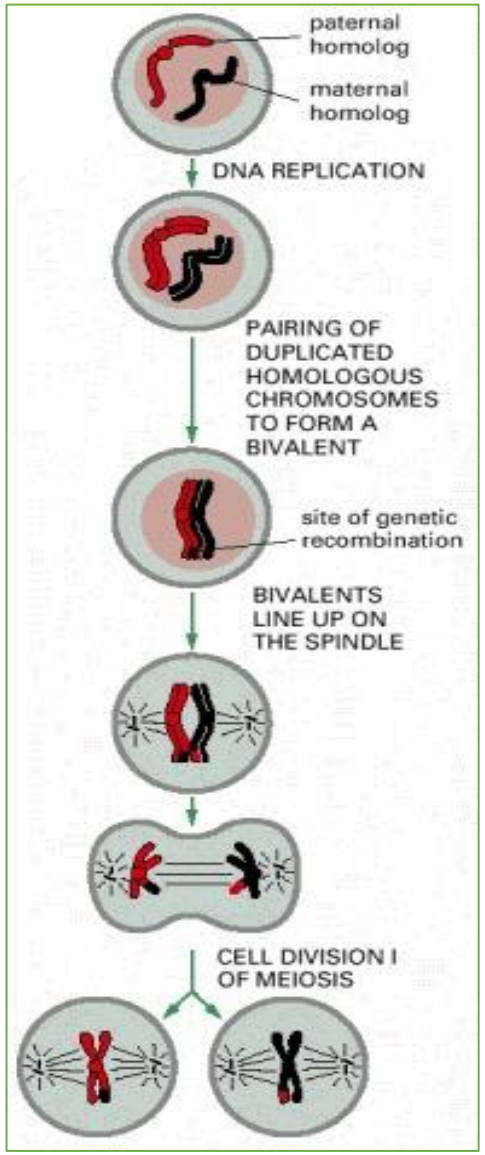
- The human life cycle begins when a haploid sperm cell fuses with a haploid ovum.
  - ✓ *The union of these gametes, culminating in the fusion of their nuclei, is* ***fertilization****.*
- **<u>The fertilized egg (zygote)</u>** is diploid because it contains two haploid sets of chromosomes bearing genes from the maternal and paternal family lines.
- As a person develops from a zygote to a sexually mature adult, mitosis generates all the somatic cells of the body.
  - ✓ Each somatic cell contains a full diploid set of chromosomes.
- **Gametes**, which develop in the gonads (testes or ovaries), are *not* produced by mitosis.
  - ✓ If gametes were produced by mitosis, the fusion of gametes would produce offspring with four sets of chromosomes after one generation, eight after a second, and so on.
- Gametes develop from specialized cells called *germ cells* in the gonads.
- <u>Gametes undergo the process of **meiosis**, in which the chromosome number is halved.</u>
  - ✓ Human sperm or ova have a haploid set of 23 different chromosomes, one from each homologous pair.
- **Fertilization restores the diploid condition by combining two haploid sets of chromosomes.**

Meiosis



- Many steps of meiosis resemble steps in mitosis.
- Both meiosis and mitosis are preceded by the duplication of chromosomes.
- In meiosis, there are two consecutive cell divisions, **meiosis I** and **meiosis II**, resulting in four daughter cells.
  - ✓ *The first division, meiosis I, separates homologous chromosomes.*
  - ✓ *The second division, meiosis II, separates sister chromatids.*
  - ✓ *The four daughter cells at the end of meiosis have only half as many chromosomes as the original parent cell.*

Meiosis



- Meiosis I is preceded by **interphase**, in which the chromosomes are duplicated to form sister chromatids.
  - ✓ *Two genetically identical sister chromatids make up one duplicated chromosome.*
  - ✓ *The sister chromatids are closely associated all along their length. This association is called sister chromatid cohesion.*
- In contrast, the two chromosomes of a homologous pair are individual chromosomes that were inherited from different parents.
  - Homologous chromosomes appear to be alike, but they may have different versions of genes, each called an *allele*, at corresponding loci.

Meiosis reduces number of chromosome sets from diploid to haploid

- **Meiosis halves the total number of chromosomes**, reducing the number of sets of chromosomes from two (diploid) to one (haploid), with each daughter cell receiving one set.
  - ✓ *Mitosis conserves the number of chromosome sets.*
- **Meiosis produces cells that differ genetically from the parent cell and from each other.**
  - ✓ *Mitosis produces daughter cells that are genetically identical* to the parent cell and to each other.

Three unique events in Meiosis

❖ **Synapsis and crossing over**

✓ *During prophase I, duplicated homologs pair up and the formation of the synaptonemal complex between them holds them in synapsis.*

✓ *Crossing over also occurs during this stage.*

✓ *Synapsis and crossing over normally do not occur during prophase of mitosis.*

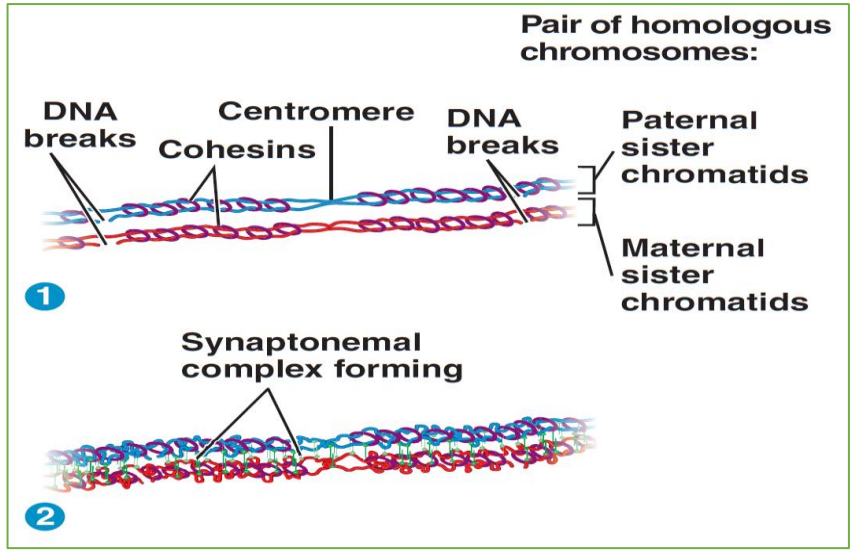
❖ **Homologs on the metaphase plate**

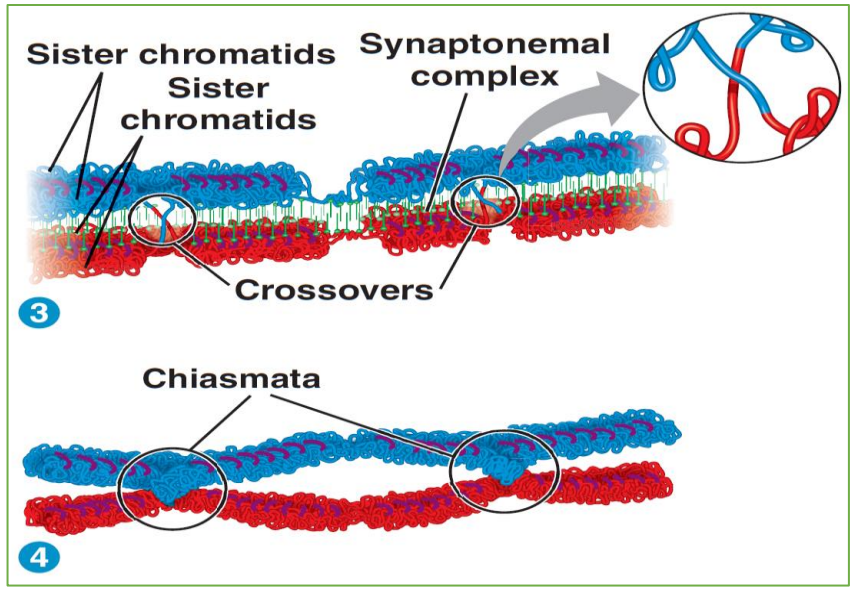
✓ ***During metaphase I** of meiosis, pairs of homologs, rather than individual chromosomes, as in mitosis, line up on the metaphase plate*

❖ **Separation of homologs**

✓ ***During anaphase I** of meiosis, the duplicated chromosomes of each homologous pair move toward opposite poles, while the sister chromatids of each duplicated chromosome remain attached.*

✓ *In the anaphase of mitosis, by contrast, the sister chromatids separate.*





Comparison of mitosis and meiosis

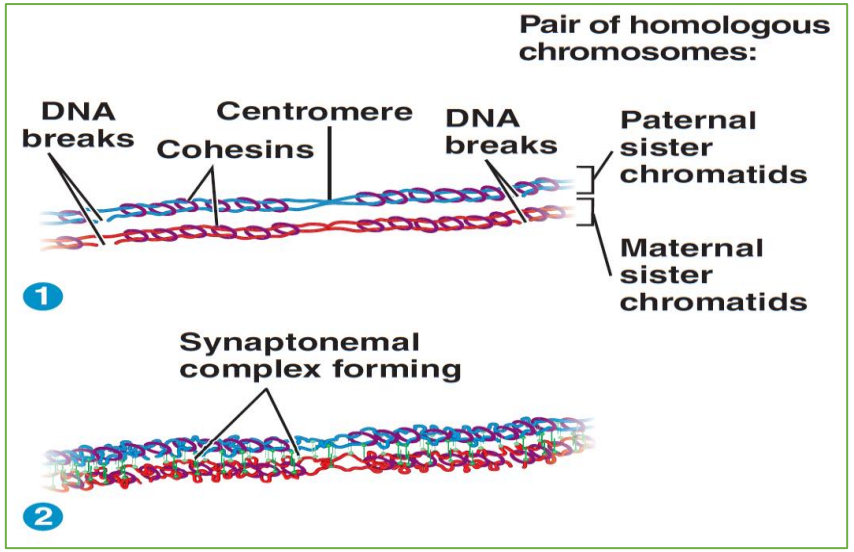
- During meiosis I, sister chromatids are attached along their lengths by protein complexes called *cohesins*.

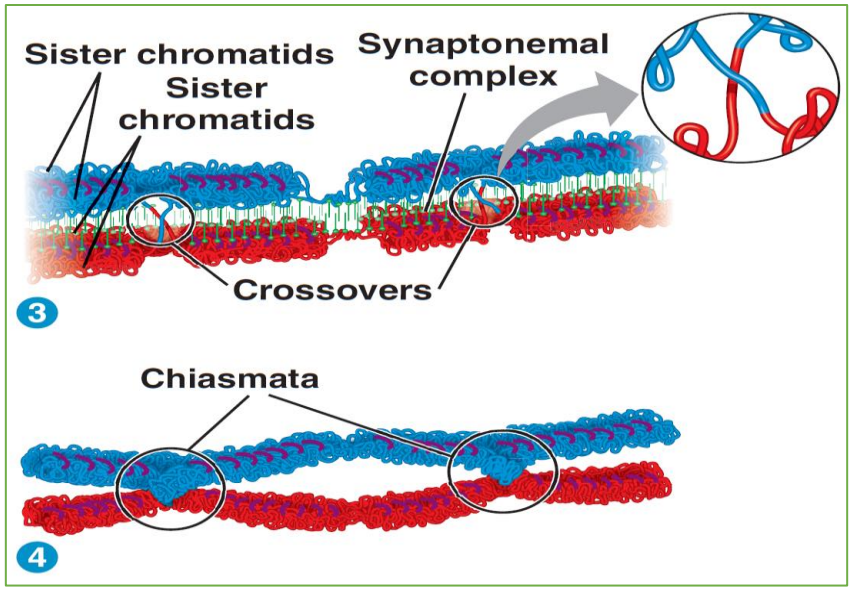
- In mitosis, enzymes remove the cohesins to allow the sister chromatids to move to opposite poles of the cell at the end of metaphase.

- In meiosis, sister chromatid cohesion is released in two steps.

1. In metaphase I, homologs are held together by cohesion between sister chromatid arms in regions beyond points of crossing over, where stretches of sister chromatids now belong to different chromosomes.
2. Both crossing over and sister chromatid cohesion along the arms result in the formation of a chiasma.

   - ✓ *Chiasmata hold homologs together as the spindle forms for the first meiotic division.*





Comparison… diploid to haploid in Meiosis-I

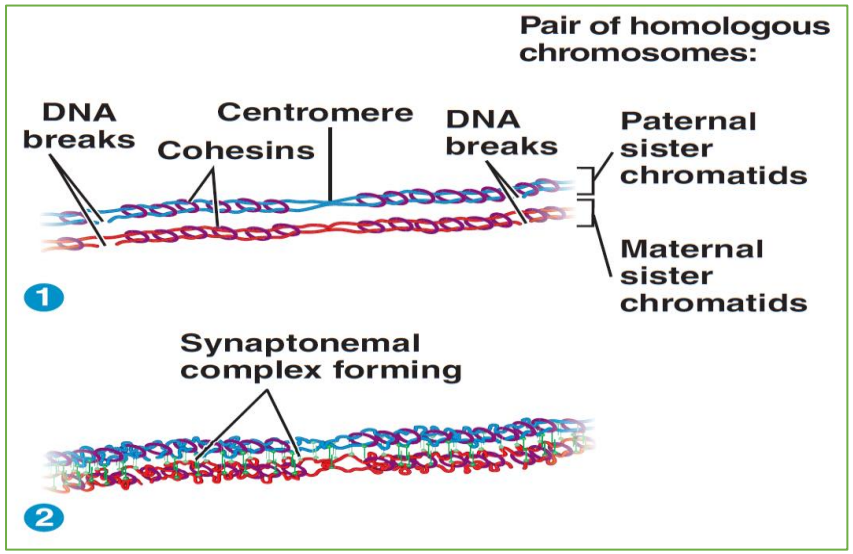
- During anaphase I, the release of cohesion along sister chromatid arms allows homologs to separate.

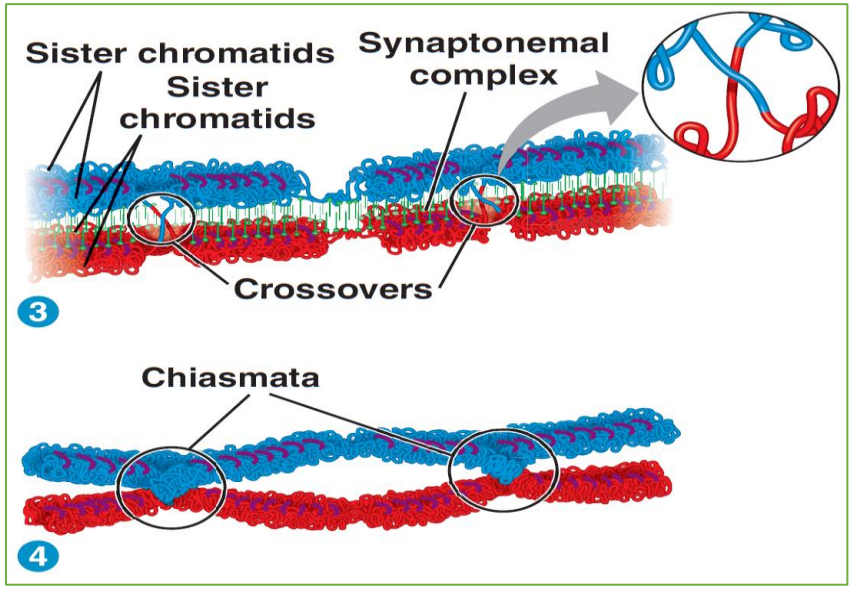
- During anaphase II, the release of sister chromatid cohesion at the centromeres allows the sister chromatids to separate.

- Thus, sister chromatid cohesion and crossing over, acting together, play an essential role in the lining up of homologous chromosomes during metaphase I.

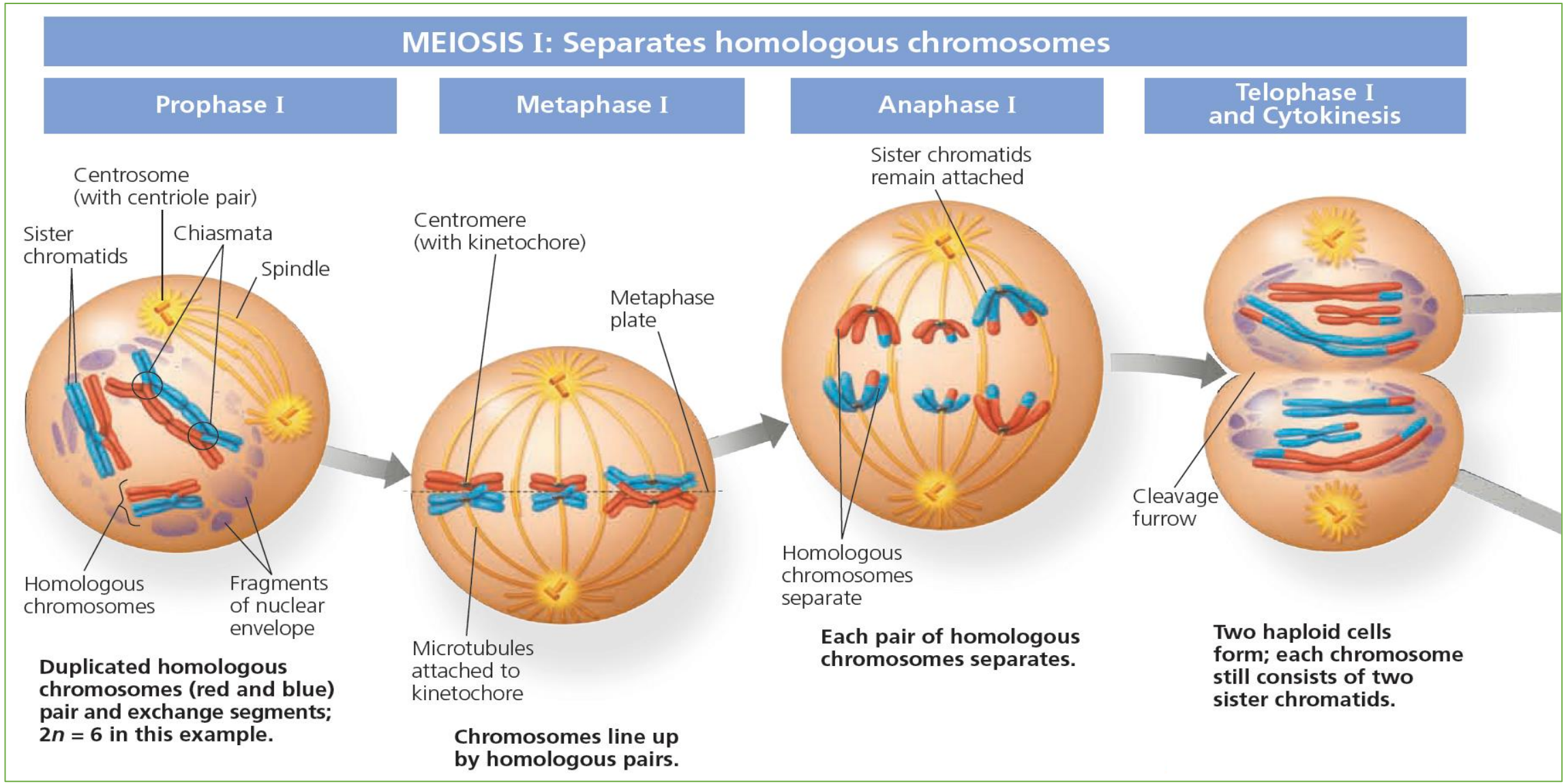
- **Meiosis I is a *reductional division* because it halves the number of chromosome sets from two (the diploid state) to one (the haploid state).**
  - ✓ ***In contrast, meiosis II is an equational division.***

- The mechanism for separating sister chromatids is virtually identical in both meiosis II and mitosis.
- The molecular basis of chromosome behaviour during meiosis continues to be a focus of intense research interest.

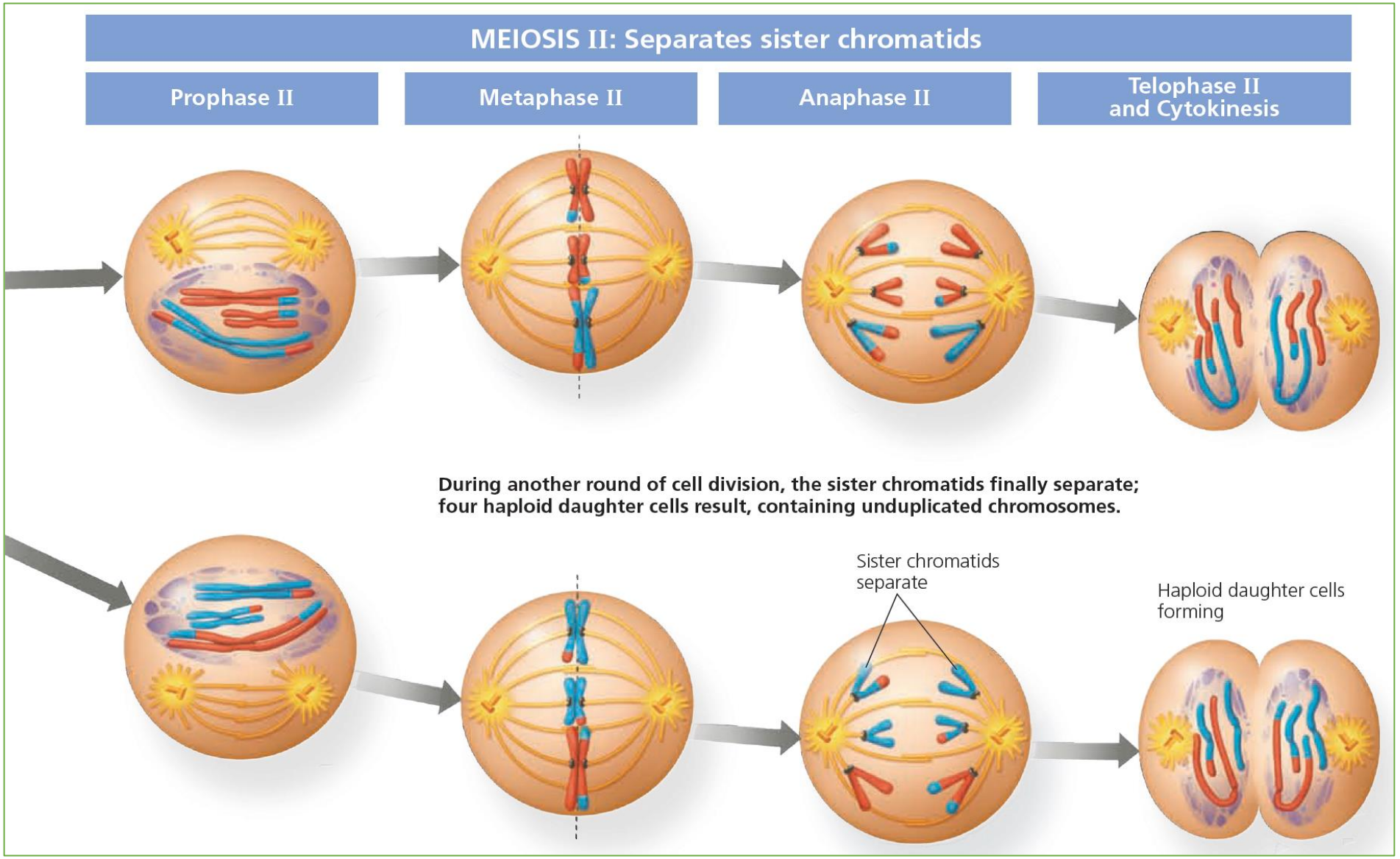




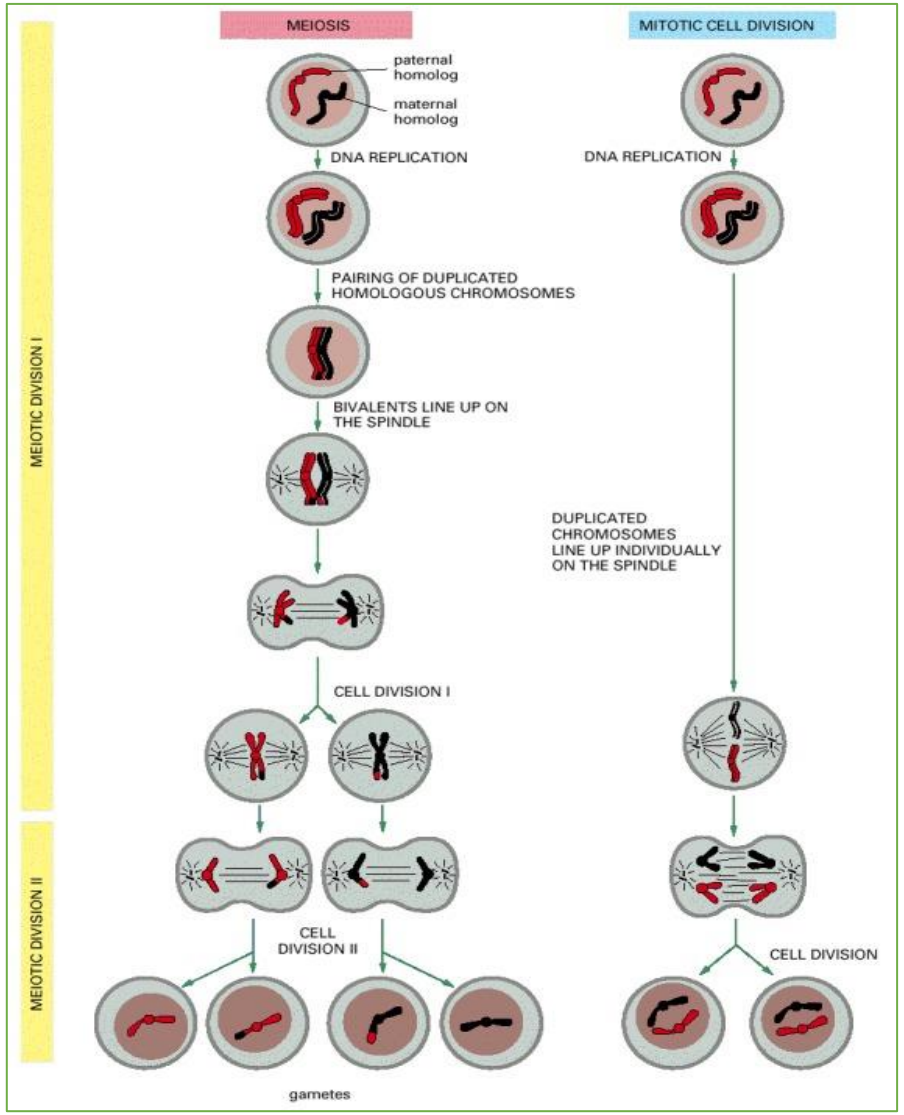
Meiosis-I and II



Meiosis-I and II



Mitosis and Meiosis



SUMMARY

Property	**Mitosis** (occurs in both diploid and haploid cells)	**Meiosis** (can only occur in diploid cells)
DNA replication	Occurs during interphase before mitosis begins	Occurs during interphase before meiosis I begins
Number of divisions	One, including prophase, prometaphase, metaphase, anaphase, and telophase	Two, each including prophase, metaphase, anaphase, and telophase
Synapsis of homologous chromosomes	Does not occur	Occurs during prophase I along with crossing over between nonsister chromatids; resulting chiasmata hold pairs together due to sister chromatid cohesion
Number of daughter cells and genetic composition	Two, each genetically identical to the parent cell, with the same number of chromosomes	Four, each haploid (*n*); genetically different from the parent cell and from each other
Role in the animal or plant body	Enables multicellular animal or plant (gametophyte or sporophyte) to arise from a single cell; produces cells for growth, repair, and, in some species, asexual reproduction; produces gametes in the gametophyte plant	Produces gametes (in animals) or spores (in the sporophyte plant); reduces number of chromosome sets by half and introduces genetic variability among the gametes or spores

Three mechanisms contribute to genetic variations

- The behavior of chromosomes during meiosis and fertilization is responsible for most of the variation that arises in each generation.

- ***Three mechanisms contribute to genetic variation arising from sexual reproduction: independent assortment of chromosomes, crossing over, and random fertilization.***

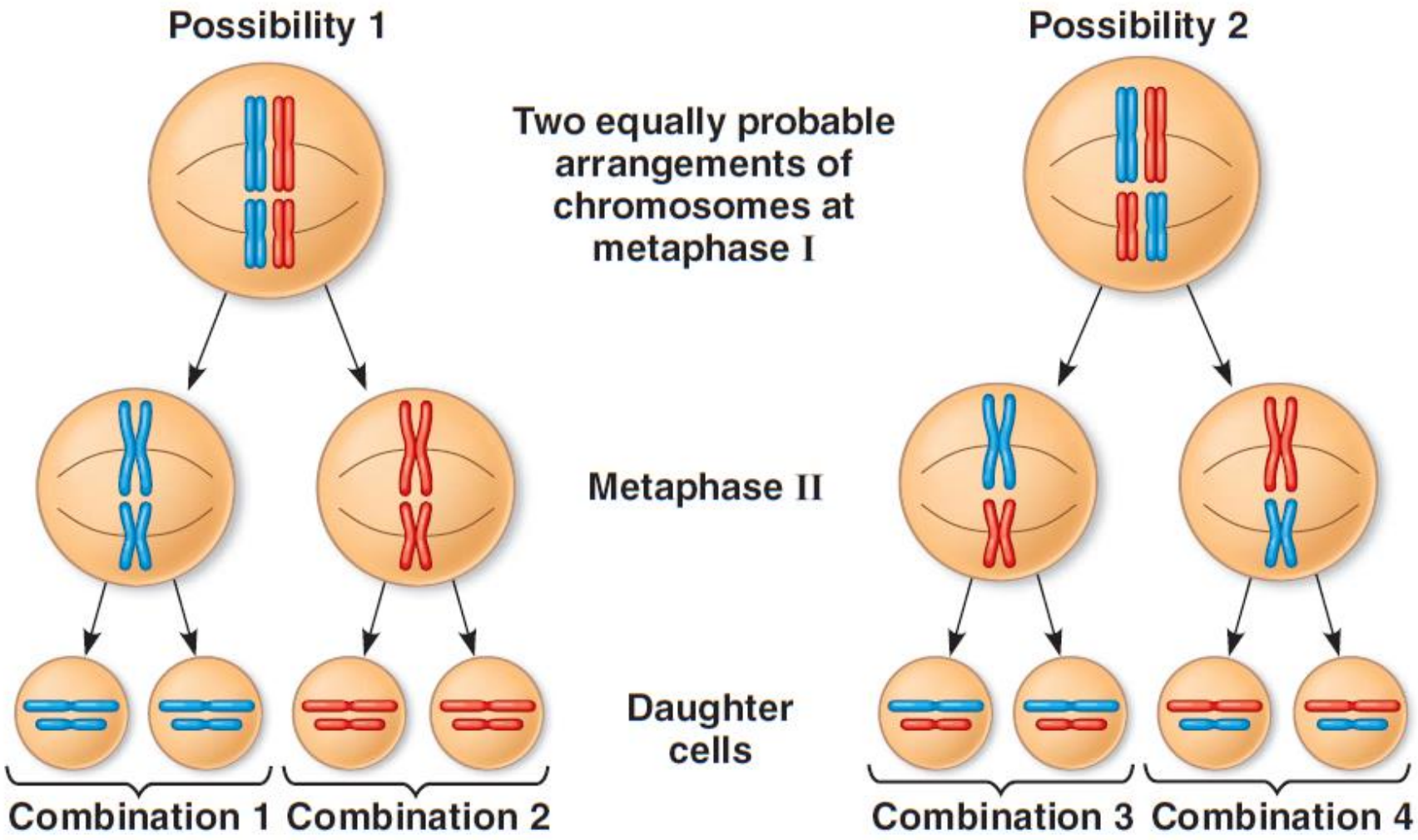
- **Independent assortment of chromosomes** contributes to genetic variability due to the random orientation of homologous pairs of chromosomes at the metaphase plate during meiosis I.

  - ✓ *There is a fifty-fifty chance that a particular daughter cell from meiosis I will get the maternal chromosome of a certain homologous pair and a fifty-fifty chance that it will receive the paternal chromosome.*
  - ✓ *Each homologous pair of chromosomes segregates independently of every other homologous pair during metaphase I.*
  - ✓ *Therefore, the first meiotic division results in independent assortment of maternal and paternal chromosomes into daughter cells*.

- The number of combinations possible when chromosomes assort independently into gametes is 2^n, where *n* is the haploid number of the organism.

  - ✓ *If n = 3, there are 2^3 = 8 possible combinations.*
  - ✓ *For humans with n = 23, there are 2^{23}, or about 8.4 million possible combinations of chromosomes*.

Independent assortment of chromosomes



Crossing over and random fertilization

- **<u>Crossing over</u>** <u>produces</u> **<u>recombinant chromosomes</u>**, which combine genes inherited from each parent.
  - ✓ *During meiosis in humans, an average of one to three crossover events occur per chromosome pair*, depending on the size of the chromosomes and the position of their centromeres.

- Crossing over, by combining DNA inherited from two parents into a single chromosome, is an important source of genetic variation.
- At metaphase II, nonidentical sister chromatids sort independently from one another, further increasing the number of genetic types of daughter cells that are formed by meiosis.

- **<u>The random nature of fertilization</u>** adds to the genetic variation arising from meiosis.

- Any sperm can fuse with any egg.

  - ✓ *The ovum is one of about 8.4 million possible chromosome combinations.*
  - ✓ *The successful sperm is one of about 8.4 million possibilities.*
  - ✓ *The resulting zygote could contain any one of more than 70 trillion possible combinations of chromosomes.*
  - ✓ *Crossing over adds even more variation to this.*

Genetic variation produced in sexual life cycles contributes to evolution

- Charles Darwin recognized **the importance of genetic variation in evolution**.
  - ✓ *A population evolves through the differential reproductive success of its variant members.*
  - ✓ *Those individuals best suited to the local environment leave the most offspring, transmitting their genes in the process.*
  - ✓ *This natural selection results in adaptation, the accumulation of favorable genetic variations in a specific environment.*
- As the environment changes, the population may survive if some members can cope effectively with the new conditions.
  - ✓ Mutations are the original source of different alleles, which are then mixed and matched during meiosis.
- New and different combinations of alleles may work better than those that previously prevailed.
  - ✓ *The ability of sexual reproduction to generate genetic diversity is one of the most commonly proposed advantages to explain the evolutionary persistence of this process.*
- In a fairly stable environment, asexual reproduction would seem to be more advantageous, because it ensures perpetuation of successful combinations of alleles.
  - ✓ Sexual reproduction is an expensive proposition—its energetic costs to the organism are higher than those of asexual reproduction.

Evolutionary adaptation of microscopic animals bdelloid rotifiers

- In spite of these apparent disadvantages, sexual reproduction is almost universal among animals as far as we know.

- While a few species are capable of **reproducing asexually under unusual circumstances**, animals that always reproduce asexually are rare.
- The best-established example, to date, is that of a group of microscopic animals called **bdelloid rotifers**

  - ✓ *This group includes about 400 species that live in a variety of environments around the world, inhabiting streams, lake bottoms, puddles, lichens, tree bark, and masses of decaying vegetation.*
  - ✓ *These animals reproduce only asexually and probably haven't engaged in sex in the 40 million years since their evolutionary origins!*

- Does the discovery of the evolutionarily successful, asexually reproducing bdelloid rotifer cast doubt on the advantage of genetic variation arising from sexual reproduction?

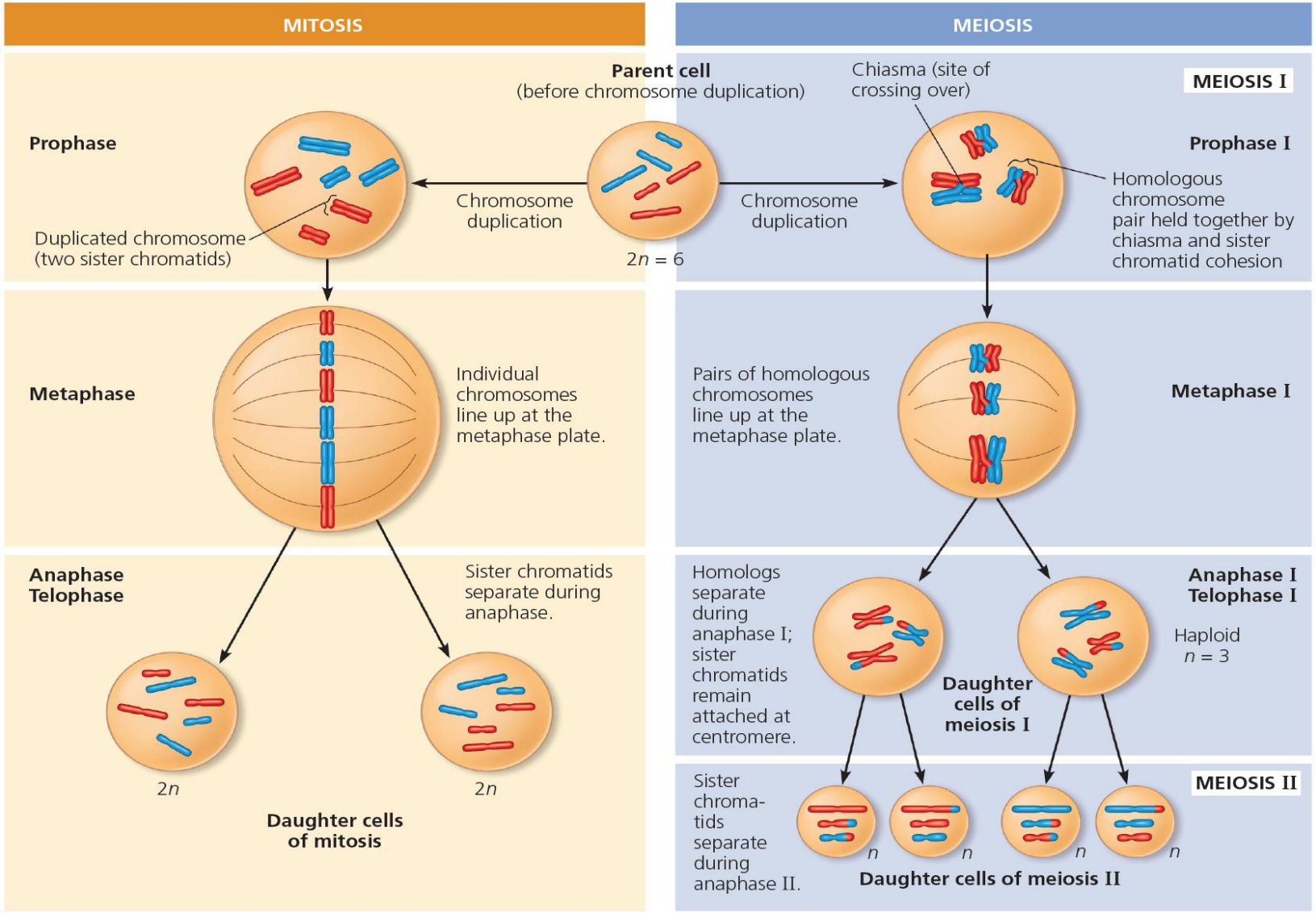
  - ✓ In fact, this group is "the exception that proves the rule."

Darwin and Mendel

Evolutionary adaptation depends on a population's genetic variation.

- **Bdelloid rotifers have mechanisms other than sexual reproduction that increase genetic diversity.**
  - ✓ *For example, they live in environments that can dry up for long periods of time, during which they can enter a state of suspended animation.*
  - ✓ *In this state, their cell membranes may crack in places, allowing entry of DNA from other rotifers and even other species.*
  - ✓ *Evidence suggests that this DNA can become incorporated into the genome of the rotifer, leading to increased genetic diversity.*
- These findings support the idea that genetic variation is evolutionarily advantageous and that a different mechanism to generate genetic variation has evolved in bdelloid rotifers.
- **In this chapter**, we have seen how sexual reproduction greatly increases the genetic variation present in a population.
- **Although Darwin realized that heritable variation makes evolution possible, he did not have a theory of inheritance.**
- Gregor Mendel, a contemporary of Darwin's, published a theory of inheritance that explained genetic variation and thus supported Darwin's theory.
  - ✓ Mendel's work was largely unknown until 1900, after both Darwin and Mendel had been dead for more than 15 years.

Summary


MITOSIS
MEIOSIS
Parent cell
(before chromosome duplication)
Chiasma (site of crossing over)
MEIOSIS I
Prophase
Prophase I
Chromosome duplication
Chromosome duplication
Homologous chromosome pair held together by chiasma and sister chromatid cohesion
Duplicated chromosome (two sister chromatids)
2n = 6
Metaphase
Individual chromosomes line up at the metaphase plate.
Pairs of homologous chromosomes line up at the metaphase plate.
Metaphase I
Anaphase
Telophase
Sister chromatids separate during anaphase.
Homologs separate during anaphase I; sister chromatids remain attached at centromere.
Anaphase I
Telophase I
Haploid
n = 3
Daughter cells of meiosis I
2n
2n
Daughter cells of mitosis
Sister chroma-tids separate during anaphase II.
MEIOSIS II
n
n
n
n
Daughter cells of meiosis II